



Consent Block

Investigating User Preferences for Smart Home Control Schemes

You are being asked to participate in a voluntary research study. The purpose of this study is to build a taxonomy of smart home user workloads and failures for both centralized (using a smart home hub) and per-device control schemes. Your responses will provide us with information regarding existing approaches to controlling smart home device infrastructures, and the reasons for opting to use certain control schemes.

Participating in this study will involve answering a set of survey questions about your experiences and current smart home setups. Participation will last approximately 20 minutes. We will not be asking you to use any of these devices in front of us, but instead we will be asking you to recount experiences with these devices. There are no foreseeable risks to you beyond those normally encountered in your daily life while participating in this survey. If you agree to participate, you will be asked to check "I consent" at

the bottom of this page indicating that you have read the following form and have been shown the goals of this study.

Principal Investigator Name and Title: Indranil Gupta

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Sponsor (if applicable): N/A

What procedures are involved?

The study procedures are an online survey that you will be asked to complete. In this survey, you will be asked about your smart home setup, including the kinds of devices you own and use, as well as how you control them. We will then ask some questions about specific experiences with these devices, and how these experiences have affected your perception of these devices and corresponding control schemes.

This research will be performed online, using an online survey software (Qualtrics). Participation will consist of one single survey, that will take approximately 20 minutes to complete.

Will my study-related information be kept confidential?

We will use all reasonable efforts to keep your personal information confidential, but we cannot guarantee absolute confidentiality. When this research is discussed or published, no one will know that you were in the study. But, when required by law or university policy, identifying information may be seen or

copied by: a) The Institutional Review Board that approves research studies; b) The Office for Protection of Research Subjects and other university departments that oversee human subjects research; c) University and state auditors responsible for oversight of research.

Will I be reimbursed for any expenses or paid for my participation in this research?

You will be compensated 4\$ USD upon completion of this survey.

Can I withdraw or be removed from the study?

If you decide to participate, you are free to withdraw your consent and discontinue participation at any time. Your participation in this research is voluntary. Your decision whether to participate, or to withdraw after beginning participation, will not affect your current or future dealings with the University of Illinois at Urbana-Champaign.

The researchers also have the right to stop your participation in this study without your consent if they believe it is in your best interests, you were to object to any future changes that may be made in the study plan.

Will data collected from me be used for any other research?

Your de-identified information could be used for future research without additional informed consent.

Who should I contact if I have questions?

If you have questions about this project, you may contact Indy Gupta at 217-265-5517 or indy@illinois.edu. If you have any questions about your rights as a participant in this study or any concerns or complaints, please contact the University of Illinois at Urbana-Champaign Office for the Protection of Research Subjects at 217-333-2670 or via email at irb@illinois.edu.

What are my rights as a research subject?

If you have any questions about your rights as a research subject, including concerns, complaints, or to offer input, you may call the Office for the Protection of Research Subjects (OPRS) at 217-333-2670 or e-mail OPRS at irb@illinois.edu. If you would like to complete a brief survey to provide OPRS feedback about your experiences as a research participant, please follow the link here or through a link on the OPRS website:

<https://oprs.research.illinois.edu/>. You will have the option to provide feedback or concerns anonymously or you may provide your name and contact information for follow-up purposes.

Please print this consent form if you would like to retain a copy for your records.

I have read and understand the above consent form. I certify that I am 18 years old or older. By clicking the “I consent” button to

enter the survey, I indicate my willingness to voluntarily take part in this study.

- ☐ I consent.
- ☐ I do not consent.

Screening Block

How many Smart Devices do you currently own? As a reminder, Smart Devices are defined as: Everyday objects made intelligent with network capabilities to execute tasks either automatically or on demand. Some examples of these devices are, but not limited to: smart plugs/switches, smart lights, smart door locks, smart thermostats, smart vacuums, window/door sensors, motion sensors, smart smoke alarms, smart cameras/security systems, smart TVs, smart speakers (amazon echo, google home), etc. Excludes smartphones, tablets, computers.

- ☐ I do not own any smart devices.
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5 or more smart devices.

Intro Block

Thank you for participating in this survey. The purpose of this survey is to understand user experiences and perceptions with smart home device control, and how these perceptions change over time. Your responses will provide valuable information to help answer our research questions. **Please keep in mind that this survey is not a test of your knowledge, but a way to learn about your experiences; there are no right or wrong answers.**

For your reference, here are some definitions to terms you may encounter in this survey:

Smart Devices: *Everyday objects made intelligent with network capabilities to execute tasks either automatically or on demand.* Some examples of these devices are, but not limited to: smart plugs/switches, smart lights, smart door locks, smart thermostats, smart vacuums, window/door sensors, motion sensors, smart smoke alarms, smart cameras/security systems, smart TVs, smart speakers (amazon echo, google home), etc. Excludes smartphones, tablets, computers.

Smart Device Control: *The ability to view and change the current state of a smart device.* The smart device itself, a physical controller or an application may be access points to control the smart device. Typically, smart devices are paired with the control medium, if that is different from the device itself, before the user can control the device. For example, you can control a smart lock on the front door either directly or with your phone.

For the purposes of this study, we will differentiate between different ways you may control your smart home devices:

Centralized Control: *Control of multiple smart devices from a central service (e.g., digital twin) or device (e.g., smart hub), that has a global view of all devices' states and access to modify the state of any and all of the devices.* All smart devices must initially be paired with the central service by the user. Some examples of centralized controllers are: Amazon Echo, Google Home mini, Apple HomePod for devices, and Amazon Alexa, Google Home, Google Assistant and Apple HomeKit for applications.

Per-device control: *Control of smart devices by pushing or rotating buttons on an individual device itself or by using an application that the device vendor has provided the customers.* The user must pair the smart device with the device

vendor if they desire controlling it by the provided app. Some examples of per-device controllers include but are not limited to: Philips Hue Bridge, Wink+GE In-Wall Smart Switch for devices, Philips Hue, ecobee for applications.

Routine: *An automation that involves smart devices. A set of actions, and potentially a set of triggering conditions, are specified. Each action changes the current state of a smart device. Each triggering condition checks the current state of a smart device or the current date and time; triggering conditions may be combined with a conjunction or a disjunction. A routine may also be manually triggered by a user, rather than triggering clauses. Other names for a routine in current products are: **recipe**, **scene**, **schedule**.*

Lastly, we define the following term(s):

Troubleshooting: *Tracing and correcting faults in a mechanical or electronic system. An example of a troubleshooting process is the following:*

1. You are at home and your smart vacuum no longer responds to your control through your Amazon Echo device.
2. You check that your smart vacuum base is connected to a power outlet; it is.

3. You check that your smart vacuum is connected to the WiFi in your home; it is.
4. You check that your smart phone has access to the internet when connected through the home WiFi; it does not.
5. You check that your home router has access to the internet; it does not.
6. You reboot your home router and wait for 5 minutes.
7. Your smart vacuum responds to your commands through your Amazon Echo device.

Please remember that you are answering on behalf of your household

Please enter your **prolific ID**.

General Smart Home Setup Questions

Which of the following smart home devices do you have powered on where you are living now? In the **first column**, select all that apply. For each owned device, in the **second column**, write how many you own and in the **third column** check each location in

the home where you have the device. Finally, in the **fourth column**, describe the device's primary use in your home.

Recall that smart devices are defined as: *"Everyday objects made intelligent with network capabilities to execute tasks either automatically or on demand."*

If you are at home, feel free to walk around and look at devices to jog your memory.

	Select	How many do you own?	Entrance Hall	Living Room	Dining Room	Kitchen	Master Bedroom	Children' room
	Check if owned	Number owned						
I do not own any smart devices	☐	0	☐	☐	☐	☐	☐	☐
Smart Plug(s) or Switch(es)	☐	0	☐	☐	☐	☐	☐	☐
Smart Light(s) (e.g. Philips Hue Bulb)	☐	0	☐	☐	☐	☐	☐	☐
Smart Door Lock(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Thermostat(s) (eg. Ecobee or Nest)	☐	0	☐	☐	☐	☐	☐	☐
Smart Vacuum/Mop(s) (e.g. Roomba)	☐	0	☐	☐	☐	☐	☐	☐
Window/Door Sensor(s)	☐	0	☐	☐	☐	☐	☐	☐
Motion Sensor(s)	☐	0	☐	☐	☐	☐	☐	☐
	Check if owned	Number owned	Entrance Hall	Living Room	Dining Room	Kitchen	Master Bedroom	Children' room
Smart Smoke Alarm(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Camera(s)	☐	0	☐	☐	☐	☐	☐	☐

	Select	How many do you own?	Entrance Hall	Living Room	Dining Room	Kitchen	Master Bedroom	Children' room
	Check if owned	Number owned						
Security System(s) (e.g. Ring, ADT, SimpliSafe)	☐	0	☐	☐	☐	☐	☐	☐
Smart TV/Casting Device(s) (e.g. Apple TV, Samsung Smart TV, Roku, Chromecast)	☐	0	☐	☐	☐	☐	☐	☐
Smart Speaker(s) (Sonos One, Amazon Echo, Google Home Mini)	☐	0	☐	☐	☐	☐	☐	☐
Smart Tag(s) (e.g., Apple AirTag, Tile Tracker)	☐	0	☐	☐	☐	☐	☐	☐
Smart Health Device(s) (e.g., smart scale, Apple watch, Fitbit health tracker)	☐	0	☐	☐	☐	☐	☐	☐
Smart Washing Machine/Dryer(s)	☐	0	☐	☐	☐	☐	☐	☐
	Check if owned	Number owned	Entrance Hall	Living Room	Dining Room	Kitchen	Master Bedroom	Children' room
Smart Dishwasher(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Fridge(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Coffee Maker(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Oven/Range(s)	☐	0	☐	☐	☐	☐	☐	☐
Smart Trash Can(s)	☐	0	☐	☐	☐	☐	☐	☐

	Select	How many do you own?	Entrance Hall	Living Room	Dining Room	Kitchen	Master Bedroom	Children's room
Other (please list device number, type, locations and main use for each device kind in the next question's text box)	☐	0	☐	☐	☐	☐	☐	☐

If you selected other, for each device:

Please list the **number** of devices, their **type**, their **location** in your home, and their **main use**.

Look at your phone. How many smart home device related *applications* do you have installed on your phone?

- ☐ 0
- ☐ 1-4
- ☐ 5-8
- ☐ More than 8

Please *list* the smart home device related applications on your phone.



When was the *last time* you purchased a smart device for your home?

- ☐ Less than one month ago
- ☐ Less than six months ago
- ☐ Less than one year ago
- ☐ More than one year ago

What was one of the *last smart devices* you purchased for your home? *Why* did you purchase it?



To what extent do you agree with this statement about the device you mentioned in the previous question: "*I was able to accomplish the goals I had in mind when I first purchased this device*"

Strongly Disagree

☐

Somewhat

disagree

☒

Neither agree nor

disagree

☒

Somewhat agree

☐

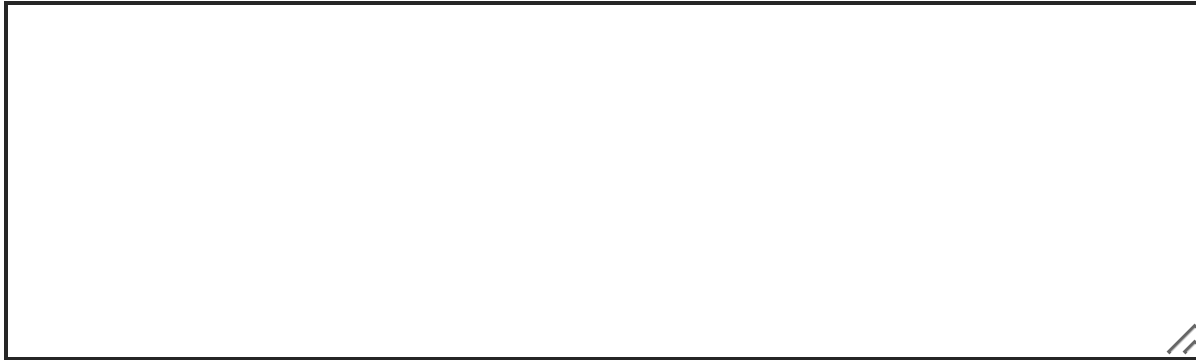
Strongly agree

☐

For any goal that you were not able to accomplish, please list the reasons.



For any goal that you were able to accomplish, please tell us how you reached them.



Attention Check (1)

Please answer the following question: What is $7+4$?

- ☐ 15
- ☐ 10
- ☐ 11
- ☐ 19

Device Control Questions

Pick a specific device in your home, and write down that device below.

The *next few questions* deal with this device that you mentioned above.

How often is the device you listed above in use (manually or via automation)?

- ☐ At least three times a week
- ☐ At least once a week
- ☐ At least once a month
- ☐ Only a couple of times since I got it

What is the way the device you listed above is *most commonly* used?

- ☐ Routines (defined as "An automation that involves smart devices. A set of actions, and potentially a set of triggering conditions, are specified. Each action changes the current state of a smart device")
- ☐ Central Control (defined as "Control of smart devices from a central service (e.g., digital twin) or device (e.g., smart hub), that has a global view of all devices' states and access to modify the state of any and all of the devices.")
- ☐ Per-Device Control (defined as "Control of smart devices by pushing or rotating buttons on an individual device itself or by using an application that the device vendor has provided the customers.")
- ☐ Voice command

☐ Content streaming/casting

Why do you choose to control the device you listed above this way(s)?



Is there any device in your home that can be controlled in multiple ways? (this can be the same device that you talked about on the previous page)

- ☐ No
- ☐ Yes
- ☐ I don't remember

If there are multiple devices that can be controlled in multiple ways, think about one of these devices.

What device is that?



How do you decide between the different options of controlling this device?



Are you aware that central smart home managers exist? Examples include Amazon Alexa, Google Home, Apple Home.

- ☐ Yes
- ☐ No

How did you learn about central smart home managers?

- ☐ Internet
- ☐ Social media
- ☐ Streaming services
- ☐ Physical/Online stores
- ☐ Word of mouth
- ☐ Other (please specify)

Device Exceptions (Blame Placement) Questions

Have you had any experience(s) when one or more of your smart devices *did not work as planned*?

- ☐ No
- ☐ Yes
- ☐ I don't remember

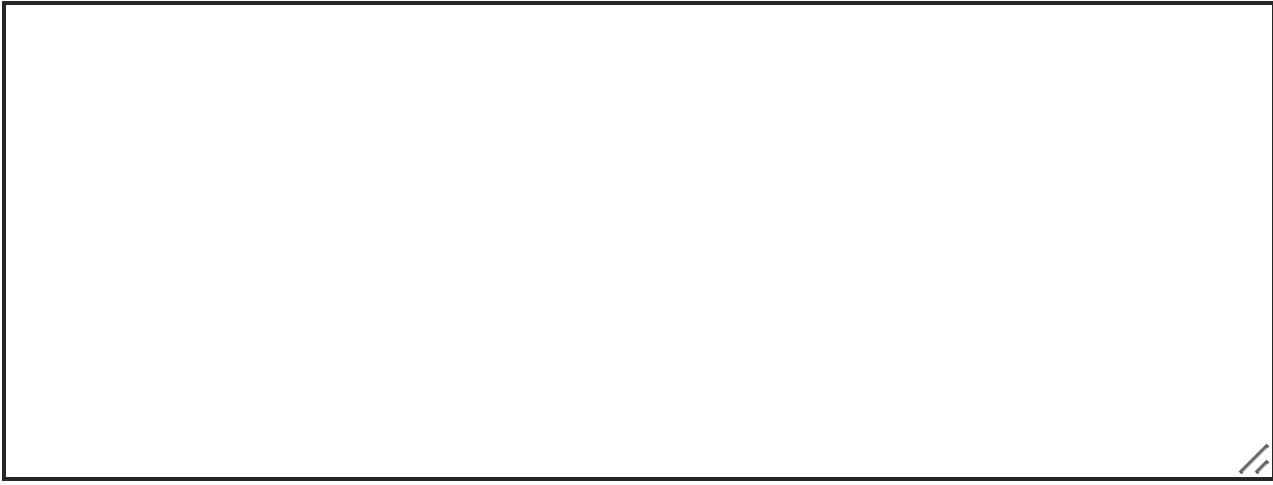
Can you describe an experience from a specific time when things didn't go as planned?



Which of the following statements represents your current experiences with smart devices?

- ☐ I am having the same kinds of issues again and again
- ☐ I encounter a specific kind of issue just once and the next ones are unrelated
- ☐ I don't remember if there are patterns among the issues I come across
- ☐ I am not encountering any issues with my smart devices

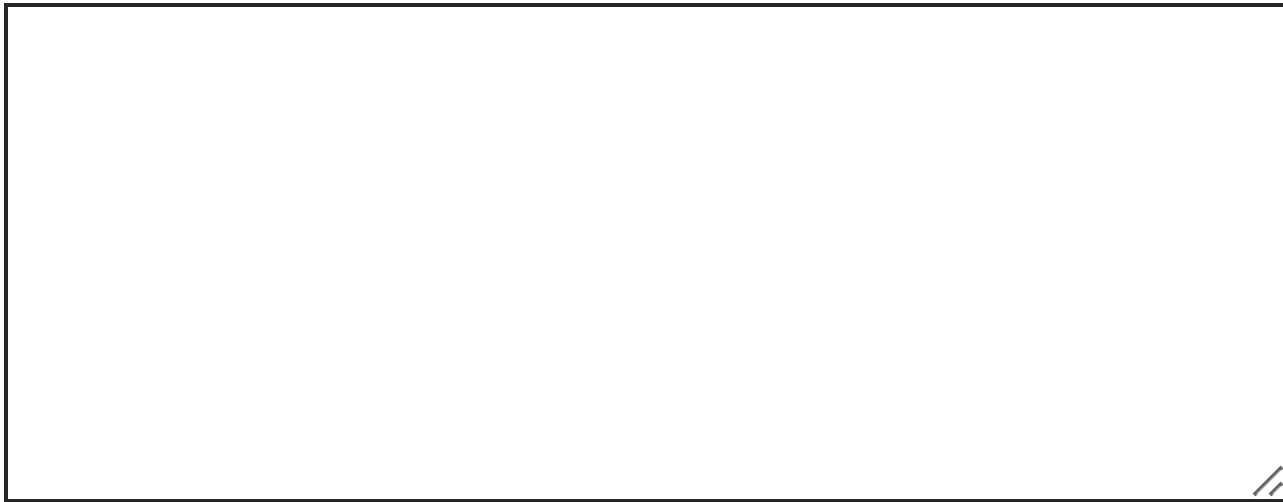
What was your reaction to any smart device failures when you started noticing them? Can you describe the reasons that caused this reaction?



Have you ever returned or stopped using a smart device because it didn't work the way you planned?

- ☐ No
- ☐ Yes
- ☐ I don't remember

What smart devices did you ever return or stop using?



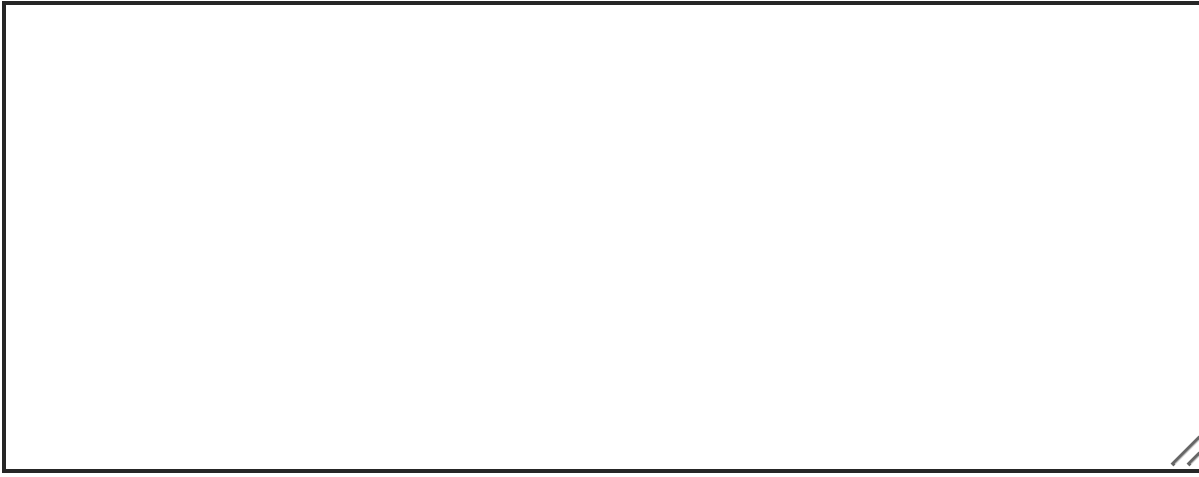
Troubleshooting Questions

How did you respond when your device did not work as expected?

- ☐ I tried to figure out what went wrong
- ☐ I tried to fix the issue/prevent it from happening again
- ☐ I stopped using the device
- ☐ I gave up on that functionality
- ☐ I don't remember
- ☐ Not applicable

Describe your troubleshooting process as a sequence of steps.

Recall that troubleshooting is defined as: *"The process of tracing and correcting faults in a mechanical or electronic system."*



Have you ever given up on using a device in your home?

- ☐ No
- ☐ Yes
- ☐ Don't remember

How did you reach that point?



Attention Check (2)

Please answer the following question: What is 15 - 11?

- ☐ 10
- ☐ 4
- ☐ 8
- ☐ 6

Questions about other spaces

Are there any more homes or spaces with smart devices that you are the main maintainer of? If so, how many?

- ☐ No
- ☐ Yes, 1 other space
- ☐ Yes, 2 or more other spaces

Can you describe where devices are placed in these spaces?

Do you think there are any similarities between the different spaces' setups?

- ☐ No
- ☐ Yes
- ☐ I don't remember

Please explain your answer to the previous question.



Demographic Block

What is your gender?

- ☐ Male
- ☐ Female
- ☐ Non-binary / third gender
- ☐ Prefer not to say
- ☐ Prefer to self-describe

How would you describe your current living situation?

- ☐ House
- ☐ Apartment
- ☐ Condominium
- ☐ Town Home
- ☐ Prefer not to say
- ☐ Other (please specify)

What is the highest degree or level of school that you have completed?

- ☐ Some high school, no diploma
- ☐ High school graduate, diploma or equivalent (for example: GED)
- ☐ Some college credit, no degree
- ☐ Associate's degree
- ☐ Bachelor's degree
- ☐ Post-graduate degree
- ☐ Other (please specify)
- ☐ Prefer not to say

What is your race?

- ☐ White
- ☐ Black or African American
- ☐ American Indian or Alaska Native
- ☐ Asian
- ☐ Native Hawaiian or Pacific Islander
- ☐ Hispanic or Latino

What is your age?

- ☐ less than 18 years old
- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55-64
- ☐ 65+

Interview Interest Block

Would you be interested in doing a follow-up interview regarding your smart home device and control experience, for additional compensation? If you indicate you are interested, we will reach out via your prolific ID.

- ☐ Yes
- ☐ No

Completion Code

Please copy and paste the following completion code into Prolific, so that we can compensate you for your time: C1DW8AKB

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